

MLink: Linking Black-Box Models for Collaborative Multi-Model Inference

Mu Yuan, Lan Zhang*, Xiang-Yang Li

University of Science and Technology of China
ym0813@mail.ustc.edu.cn, zhanglan@ustc.edu.cn, xiangyangli@ustc.edu.cn

Abstract

The cost efficiency of model inference is critical to real-world machine learning (ML) applications, especially for delay-sensitive tasks and resource-limited devices. A typical dilemma is: in order to provide complex intelligent services (e.g. smart city), we need inference results of multiple ML models, but the cost budget (e.g. GPU memory) is not enough to run all of them. In this work, we study underlying relationships among black-box ML models and propose a novel learning task: model linking. Model linking aims to bridge the knowledge of different black-box models by learning mappings (dubbed model links) between their output spaces. Based on model links, we developed a scheduling algorithm, named *MLink*. Through collaborative multi-model inference enabled by model links, *MLink* can improve the accuracy of obtained inference results under the cost budget. We evaluated *MLink* on a multi-modal dataset with seven different ML models and two real-world video analytics systems with six ML models and 3,264 hours of video. Experimental results show that our proposed model links can be effectively built among various black-box models. Under the budget of GPU memory, *MLink* can save 66.7% inference computations while preserving 94% inference accuracy, which outperforms multi-task learning, deep reinforcement learning-based scheduler and frame filtering baselines.

Introduction

Multi-model inference workloads are increasingly prevalent, e.g., smart speaker assistants (Bentley et al. 2018), smart cities (Duan et al. 2018), drone-based video monitoring (Dilshad et al. 2020), multi-modal autonomous driving (Feng et al. 2020), etc. Besides the accuracy of the trained models, costs in the inference phase can become the bottleneck to the quality of services, especially for delay-sensitive tasks and resource-limited devices.

Towards cost-efficient inference, existing work explored various perspectives to achieve the resource-performance trade-offs. Multi-task learning and zipping (He, Zhou, and Thiele 2018; Sanh, Wolf, and Ruder 2019; Crawshaw 2020; Zhang and Yang 2021) can reduce the computing overheads by sharing neurons among different tasks; Model compression (Hinton, Vinyals, and Dean 2015; Liu et al. 2018; Gold-

blum et al. 2020; Bai et al. 2020) techniques attempt to eliminate parameters and connections not related to the inference accuracy; Inference reusing (Guo et al. 2018; Ning, Guan, and Shen 2019) approaches aim to avoid the same or similar computations; Source filtering (Li et al. 2020) methods try to transmit only necessary input data to backend ML models. Adaptive configuration (Jiang et al. 2018) and multi-model scheduling (Yuan et al. 2020a) were proposed to make inference workloads adaptive to the dynamics of input content. We summarize them as answers to an interesting question:

How to obtain as accurate inference results as possible without the exact execution of ML models?

From this perspective, multi-task learning and model compression generates a lighter model for the same inference task(s) by pruning the original model(s). Inference reusing and source filtering techniques reuse previous inference results as the predicted results through analyzing the correlation between inputs. Based on the observation that, for some input data, the accuracy of expensive and cheap models is similar, adaptive configuration analyzes the input dynamics and predicts the inference results of expensive models by executing cheap ones. Adaptive multi-model scheduling predicts unnecessary inference results as empty using the executed models' outputs as the hint information.

We address this problem from a novel perspective: *linking black-box models*. We were motivated by the insight that even ML models that are different in input modalities, learning tasks, architectures, etc., can share knowledge with each other, since ML models are prone to "overlearning" (Song and Shmatikov 2020) and outputs of different models have semantic correlations (Tan et al. 2018). If we can effectively bridge the knowledge among ML models, we can directly predict inference results of remaining models based on executed models' outputs. If the cost of this prediction is low, it is promising to improve the resulting accuracy of inference results from all models under a limited cost budget, compared to the original workflow where results of unexecuted models cannot be obtained at all. To realize this vision, the following two main challenges need to be solved:

(1) **How to build knowledge-level links among black-box and highly different ML models?** In practice, deployed ML models could have different architectures and input modalities, and they could be developed by different programming languages and ML frameworks. The hetero-

*Lan Zhang is the corresponding author.

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